



Exploring the validity of the ASQ-SE for socio-emotional competency screening of a low-risk Asian cohort at 2 years of age

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ABSTRACT

Aims: To assess the Ages & Stages Questionnaire: Social-Emotional (ASQ-SE)'s concurrent validity in a low-risk Singapore cohort and study its association with maternal mental health status.

Methods: Concurrent validity of the parent-filled ASQ-SE with Child Behavior Checklist (CBCL1.5–5) was evaluated in 341 children at age 24 months. Data on maternal anxiety and depression were collected using the State-Trait Anxiety Inventory (STAI) and Beck Depression Inventory-Second Version (BDI-II). ASQ-SE cut-off scores based on receiver operating characteristic curve were compared to CBCL scores to derive a local ASQ-SE "at risk" cut-off score. Correlations of ASQ-SE with CBCL scores and with maternal STAI and BDI scores were evaluated using Pearson coefficients.

Results: Using a cut-off score of 51 at 24 months, ASQ-SE had acceptable concurrent validity, with an AUC of 0.819(0.765–0.872), 70 % sensitivity and 79 % specificity. Mothers of children with "at-risk" ASQ-SE scores had significantly higher STAI and BDI-II scores. ASQ-SE had moderate- high correlations ($r = 0.32$ – 0.53) ($p < .01$) with CBCL scores at 24 and 48 months and with maternal mental health status($r = 0.32$).

Interpretation: ASQ-SE can be a useful tool for screening child's socio-emotional competence for primary health care use in Singapore Dyadic mental health screening would be helpful in identifying families at risk.

1. Introduction

The early childhood years are critical in fostering healthy development in the domains of cognitive skills, social-emotional competence and subsequent adult mental health [1,2]. With a greater understanding of the ecobiodevelopmental framework of child development, there is increasing awareness and recognition of the importance of these early years [3].

Difficulties in socio-emotional competence or mental health

concerns are seen in 11–20 % of young children between birth to 5 years and may impact long-term well-being and learning [4]. If mental health problems in the early years are missed they may eventually become disorders and negatively affect school success, adolescent behavior and mental health [5] with possible socio-economic costs to both the family and community. Furthermore, 23–61 % of children with a diagnosis of behavioral or emotional disorder will also have a similar diagnosis in the future [4].

Among risk factors associated with disruptions to a child's social-

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emotional development, maternal perinatal mental health concerns have been associated with compromised behavioral, cognitive, and social-emotional outcomes [6]. Depression and anxiety, the commonest pregnancy related mental health concerns with an estimated prevalence of 12–20 % have shown detrimental effects on attachment, early development and socio emotional competence in childhood and adolescence [7]. Thus, timely identification and support for the mother-child dyad mental health concerns are important to promote optimal outcomes into school age, adolescence and beyond [8].

1.1. Early screening of social-emotional competencies

The increasingly accepted international consensus that the early years are the best time to identify and support children at risk of later mental health concerns makes, regular mental health screening through the preschool, school and adolescent years, important [9] [5] [3].

However, identification and intervention for social-emotional difficulties have lagged behind screening in other developmental areas. Fewer than 1 in 3 children with mental health problems are identified in primary care settings and fewer than 1 in 8 children with identified problems receive treatment [4] [9].

US data has shown that 50 % of pediatric providers report never having used any formal measure to assess mental health and fewer than 7 % used a standardized measure during 50 % or more of visits [10].

1.2. Tools for social-emotional screening

Validated parent-completed questionnaires would enable improved detection of socio-emotional problems in preschool children at primary health-care level as clinical judgment alone has shown limited sensitivity [4].

A systematic review by Pontoppidan et-al and the AAP [11] [12] have listed commonly used parent-reported tools to assess socio-emotional competency and problems including the Ages and Stages Questionnaires-Social-Emotional (ASQ- SE), and the Child Behavior Checklist (CBCL 1.5–5). In view of its high validity and reliability, the CBCL [13] is often used as the gold standard or criterion measure to evaluate socio-emotional competence. However it may not be practical to use in routine screening since it has a large number of items and has a deficit based approach which is not parent friendly and may be a better fit in clinical settings [12]. ASQ-SE [14] is significantly shorter than the CBCL with between 19 and 33 items at different ages, assesses behavioral repertoire more broadly, and applies a strength-based approach more suitable in primary care settings [12].

ASQ-SE has shown adequate psychometric properties in the US [15] but data in Asian settings from children below 2 years is only emerging. Data from Asian countries has found promising psychometric findings of the ASQ:SE, Second Edition (ASQ:SE-2) in children from 42 to 54 months in Taiwan [16]. A study [17] on a representative sample of 2830 children in China reported that the ASQ:SE-2 classification results (i.e., typical, at risk) reached a 85 % agreement with the results on convergent measures including the Chinese Infant-Toddler Social and Emotional Assessment and the Chinese CBCL scales [18,19]. Also in the 2022 study [17], the two-factor structure (i.e., social and emotional) of the ASQ:SE-2, was found to be a poor fit for the 2-month age interval, but a good fit for older age intervals. In Singapore, there is currently no routine screening for socio-emotional competency from 0 to 5 years of age with a paucity of data on the prevalence of socio-emotional difficulties in this important age group.

The aims of this study were to assess the psychometric properties of the ASQ-SE, in a low-risk Singapore cohort by examining its internal consistency and concurrent validity with the CBCL and also to study the relationship between ASQ:SE scores and maternal mental health status.

2. Methods

2.1. Participants

The current study was part of a larger parent-offspring longitudinal cohort research project, the Growing Up in Singapore Towards Healthy Outcomes (GUSTO) study [20]. GUSTO recruited pregnant women at Singapore's two major public maternity units between June 2009–September 2010 and collected data on early life growth and developmental outcomes through the antenatal period and 0–3 years in participating children. Data was collected from six home and three clinic visits. The current sub-study, recruited 341 participants from the larger GUSTO cohort.

The GUSTO cohort study was approved by the National Healthcare Group Domain Specific Review Board and the SingHealth Centralized Institutional Review Board and written informed parental consent was obtained.

2.2. Measures

2.2.1. Ages & stages questionnaires-SE [14]

The ASQ-SE, a norm referenced screening tool, evaluates socio-emotional competency and problems for ages between 3 and 66 months. Each scoring item is rated with one of three possible options: “rarely or never”, “sometimes”, and “most of the time” and then combined into a total score for comparison against normative cut-off scores. Scores above the cut-off indicate the need for referral and further mental health assessment.

In the current study, parents/caregivers completed the 24-month ASQ-SE with 26 scoring items and total scores ranging from 0 to 390.

In Singapore's multi-linguistic context, the ASQ:SE was translated into Mandarin, Malay, and Tamil. The translations followed a forward and backward procedure to improve linguistic equivalence [21]. The translations were developed by psychology or related science graduate students with a bachelor's degree or higher. Among the 341 returned questionnaires, 339 (99.4 %) were completed in the original English, 1 (0.5 %) was completed in translated Mandarin version, 0 (0 %) in Malay, and 1 (0.5 %) in Tamil. The sample sizes in non-English groups (i.e., Mandarin, Malay, and Tamil) were not sufficient for analyses identifying differential item functioning across different language versions.

2.2.2. Child behavior checklist for ages one and a half- to five-years old (CBCL 1.5–5)

The 99-item CBCL computes Total, Internalizing and Externalizing problem scores based on parental responses of their child's behavioral/emotional problems. Internalizing Problems consists of withdrawn/depressed and anxious syndrome scales while externalizing problems consist of the oppositional, aggressive behavior and overactive syndrome scales.

Each item was scored on a 3-point scale (0 = “not true (as far as you know)”, 1 = “somewhat true” and 2 = “very true or often true”) with total scores ranging from 0 to 198. In this study, parents completed the CBCL at 24 and 48 months.

2.2.3. State-trait anxiety inventory [22]

The STAI is a 40-item scale comprising of 2 subscales, the 20-item State Anxiety Scale (S-Anxiety) and a 20-item Trait Anxiety Scale (T-Anxiety). We used the S-Anxiety scale to measure caregivers' current state of anxiety at child's age of 24 months.

Each item is scored on a 4-point Likert scale with total scores ranging from 20 to 80 and higher scores indicating greater anxiety. Scores above 40 were used as the cut-off to indicate high maternal state-anxiety.

2.2.4. Beck Depression Inventory-Second version(BDI-II) [23]

The BDI-II is a 21-item self-administered survey, divided into 2 subscales, affective symptoms and somatic symptoms to assess presence

and intensity of symptoms of depression. Items are scored on a scale of 0–3 in increasing order of severity with cut-off scores of 14–19, 20–28 and 29–63 indicating mild, moderate and severe symptoms, respectively. Maternal BDI-II data was collected at child's age of 24 months and a cut-off score of 13 was used in this study to indicate maternal depression.

2.3. Procedures

2.3.1. Demographic data collection

GUSTO data collection was described in a previous publication [20]. For this sub-study demographic variables analysed included ethnicity, maternal educational level, and child's gender.

2.3.2. ASQ-SE data

The 24-month ASQ-SE questionnaire with an explanatory cover letter and instructions was mailed or handed to caregivers at the 24-month visit and then similarly returned.

2.4. Data analyses

Total ASQ-SE scores were reported as descriptive values in the form of numbers, medians and interquartile ranges. Two different methods were used to generate ASQ-SE cut-off scores as recommended in the User's Guide [14]. The first was using a semi-interquartile range method (median + 1.5*Semi IQR) and the second using receiver operating characteristic(ROC) curve method in which CBCL scores at 24 months were used as a criterion measure. CBCL cut-off scores were generated based at the 98th centile for the syndrome scales and on 90th percentile for broadband scales (internalizing, externalizing and total problems) and these were used to generate the ROC curve. Subsequent analyses were performed by comparing children's CBCL and ASQ:SE data to examine sensitivity, specificity, positive and negative predictive values of the ASQ: SE.

Pearson correlation coefficients between ASQ-SE 24 m and CBCL scores were used both to evaluate the criterion validity of the ASQ:SE and to examine the relationship between ASQ-SE and maternal STAI and BDI scores. Correlations >0.50 were considered large, 0.30–0.50 were moderate, and <0.30 indicated a small magnitude.

Analysis was performed using IBM SPSS version 26.

3. Results

3.1. Participants

A total of 341 valid ASQ-SE responses were collected in the study. Table 1 shows the cohort's descriptive demographic data along with ASQ-SE and CBCL data and maternal mental health screening results at

Table 1
Demographic characteristics of children and families in current study ($n = 341$) results of the child and maternal measures.

Characteristics	
Child gender n (%)	
Male	177 (51.91)
Female	164 (48.09)
Ethnicity n (%)	
Chinese	176 (51.61)
Malay	100 (29.33)
Indian	58 (17.01)
Maternal education n (%)	
≤High School	70 (20.83)
College and above	266 (79.17)
Median ASQ:SE score	36.40
CBCL Total Problem Syndrome scores mean (SD)	33.36 (20.75)
Mean maternal STAI – State Anxiety scale scores (SD)	34.33 (10.12)
Mean maternal BDI scores (SD)	7.20 (7.34)

2 years. As listed in Table 1, 177 (49.30 %) were boys, 51.61 % of the participating children were of Chinese ethnicity, followed by 29.33 % Malay and 17.01 % Indian ethnicity.

3.1.1. Establishing the local Singapore Cut-off for 24-month ASQ:SE-2

As ASQ-SE total scores of the 341 participants showed a skewed distribution (Fig. 1) with scores ranging from 25 to 60 at the 25th and 75th centile respectively and the highest score being 135 t, means and standard deviations were not used for identifying the cut-off score.

Cut-off scores for our cohort at 24 months of age based on Median + 1.5*Semi IQR, and ROC cut-off values were 36.4, 62.6 and 51, respectively compared to 28.4, 54 and 50 for the US samples. ASQ-SE's internal consistency measured by Cronbach's coefficient alpha was 0.68. The cut-off score of 62.6 based on the Median + 1.5*Semi IQR was much higher than both the ROC based cut-off of 51 and the US cut-off scores. The ROC based cut-off of 51.00 was thus selected for this study.

Using the locally generated ROC cut-off based on CBCL as the criterion measure, 215(63 %) of the 341 study children were found to be "typical" on both ASQ-SE and CBCL while 49(14 %) of the children were "at risk" on both measures. Of the remaining, 21(6 %) had "typical" ASQ-SE scores but were found to be at risk on the CBCL, while 56 (16 %) were at risk on the ASQ-SE but screened normal on CBCL.

Thus, 236 (69 %) of the study children were "typical" on the ASQ-SE while 105 (31 %) were considered "at risk".

3.1.2. Converging validity and predictive validity of the ASQ:SE

To examine our cut-off score's validity, the percent of agreement, sensitivity, specificity, under and over identification rates, positive predictive value (PPV), and negative predictive value (NPV) were calculated by comparing the screening results using the local ROC cut-off with the CBCL results. As Table 2 shows, percentage agreement between the ASQ:SE and CBCL was 77.4 % with an Area under the curve (AUC) of 0.82. Both sensitivity and specificity were ≥ 70 %. Only 6.2 % of participants were under-identified. The PPV was only 46.7 % but NPV was above 90 %.

In addition, ASQ:SE scores had moderate-high correlations with both CBCL total problem scores (CBCL-TPS) and subscales at 24 and 48 months as seen in Table 3. All comparison dyads showed correlations significant at $p < .01$, with magnitudes ranging from moderate (e.g., $r = 0.327$, $p < .01$) to high (e.g., $r = 0.53$, $p < .01$) levels except for a small correlation of 0.288 for sleep problem syndrome scale at 48 months. At 24 months, the highest correlations (>0.5) of the ASQ-SE were seen for the Internal problem syndrome scale and total CBCL scores, while the lowest correlations of 0.35–0.37 were seen for the attention, anxious/depressed and emotional reactive syndrome scales.

3.1.3. Association between ASQ:SE scores and maternal mental health parameters/status at 24 months

Table 4 shows CBCL and maternal STAI and BDI Scores in children who were screened "typical" or "at risk" on the ASQ-SE with significant differences in scores for all outcome measures ($p < .001$).

Comparison of maternal anxiety traits and depression scores revealed that 47.6 % of mothers whose children had elevated ASQ-SE scores had raised STAI scores and 26.6 % had elevated BDI scores with both being significantly higher than in mothers whose children had normal ASQ-SE scores. Significant medium-level correlations were found between ASQ:SE and STAI ($r = 0.312$, $p < .01$) as well as with BDI-II ($r = 0.326$, $p < .01$).

4. Discussion

Systematic reviews [12] have demonstrated the critical need for measuring preschool socio-emotional development at primary health-care level in order to accurately identify children at risk and to facilitate early evaluation and intervention.

With no local data available, it was important to validate currently

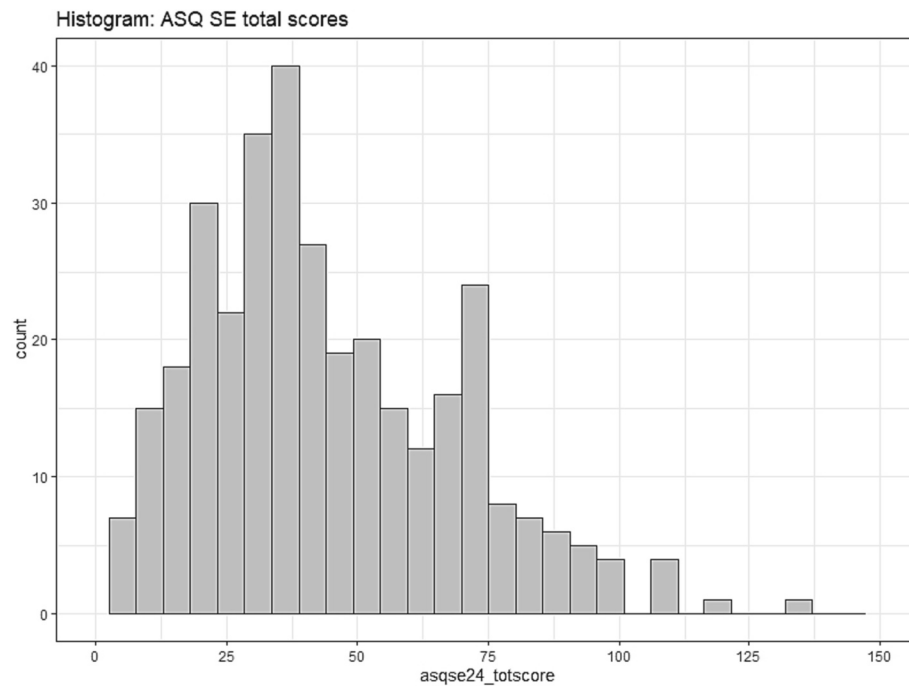


Fig. 1. ASQ:SE total scores by number of children for the 24-month questionnaire (n = 341).

Table 2

Agreement between ASQ:SE classification and CBCL classification (n = 341).

	Using local ROC cutoff score(%)
Percent agreement	77.4
Sensitivity	70.00
Specificity	79.3
Under-identified	6.2
Over-identified	16.4
Positive predictive value	46.7
Negative predictive value	91.1
Area under the curve	0.819 (0.765–0.872)

Table 3

Correlations between ASQ:SE 24-month and CBCL all domains and externalizing, internalizing, Total (n = 341).

	Correlation of ASQ-SE at 24 months and CBCL at 24 months	Correlation of ASQ-SE at 24 months and CBCL at 48 months
CBCL	Correlation with ASQSE	
Emotional Reactive syndrome scale	0.37**	0.34**
Anxious Depressed syndrome scale	0.37**	0.33**
Somatic Complaints syndrome scale	0.44**	0.35**
Withdrawn syndrome scale	0.46**	0.41**
Internal Problem syndrome scale	0.51**	0.43**
Attention syndrome scale	0.35**	0.34**
Aggressive Problem syndrome scale	0.46**	0.45**
External Problem syndrome scale	0.46**	0.45**
Sleep Problem syndrome scale	0.36**	0.29**
CBCL Total problem score	0.53**	0.47**

* $p < .05$; ** $p < .01$; *** $p < .001$.

Table 4

Comparison of CBCL and maternal Mental health status in children with typical and at risk ASQ-SE scores.

	ASQ:SE 24 month Typical (n = 236)	ASQ:SE 24 month At risk (n = 105)	Significance of group difference
Mean (SD) maternal STAI State 24-month scores	32.29 (9.37),	38.90 (10.29),	***
Mean (SD) maternal BDI-II 24-month scores	5.66 (5.91),	10.67 (8.94), n = 100	***
Raised STAI State scores >40-n(%)	42 (17.7 %)	50 (47.6 %)	***
Raised BDI scores >14 - n (%)	23 (9.7 %)	28 (26.6 %)	***

*** $p < .001$.

available screeners within the local population.

Our study evaluated the reliability and validity of the ASQ-SE in a low-risk Singapore cohort at 24-months of age and demonstrated that local cut-off scores at 24 months had acceptable psychometric properties, with adequate sensitivity of 70 %, specificity of 79 %, and an AUC of 0.819(0.765–0.872) when using CBCL as the criterion measure. Our findings were similar to the findings of >70 % sensitivity and specificity between 0 and 3 years reported by Alvarez-Nunez et al. from Spain [24].

Significant correlations at medium-high levels were found between ASQ-SE and the CBCL -TPS and subscales supporting the concurrent validity of AS-SE. The original ASQ-SE [14] manual reported sensitivity similar to our study's findings of 70 %, while our specificity was lower than the manual's 93 % specificity. Future research is needed to further explore possible cultural differences in our population, as these may affect socio-emotional function and norms. Thus possibly resulting in our lower specificity PPVs in our study were relatively low at 46.6 % and also seen in other studies [25]. The higher NPV and specificity compared to sensitivity and PPV seen in our study would imply that normal ASQ-SE screening results could be used to reassure parents about normal SED of their children. Anticipatory guidance should continue to be provided about appropriate preventive care with ongoing surveillance

and age-appropriate screening for emotional health.

Correlations between ASQ-SE and CBCL-TPS were similar to those reported in a Norwegian study using CBCL- TRF (0.49,95%CI 0.33 to 0.63) and in a recent Spanish study [24] [25] Another study in a Dutch primary health care population showed low discriminatory capacity at six and fourteen months of age but improved subsequently at 24 months with sensitivity of 0.66, specificity of 0.91, an AUC of 0.88 and internal consistency of 0.62 when using the 90th centile of the CBCL- TPS [26] Elevated CBCL-TPS scores in that study were seen in 9.8 % of the children. In comparison 6.6 % of our cohort had elevated CBCL-TPS scores with an AUC of 0.81 and higher sensitivity and internal consistency but lower specificity.

Our reporting of ASQ-SE's sensitivity and specificity using the CBCL as the criterion measure also resonated with findings from systematic reviews, demonstrating lower sensitivity (0.68) and specificities (0.84) for studies using alternative cut off scores developed based on the local population The reviews also documented ASQ-SE's increasing reliability from 24 months of age onwards [27,28].

As often discussed, a trade-off exists between sensitivity and specificity when evaluating screening tools. Specificity assesses a test's ability to correctly identify individuals without concerns, while sensitivity assesses the ability to correctly identify individuals with concerns. As a screening measure, sensitivity may be traded off for specificity as low specificity would yield large number of false-positive results with ensuing unnecessary referrals. This would limit upscaling and generalizability of the screening test in low-risk populations.

We chose a cut-off score acceptable for developmental screening for both sensitivity and specificity as this will enable identifying children with subthreshold symptoms who may benefit from closer monitoring.

4.1. Predictive validity with CBCL at 48 months

The usefulness of socio-emotional competence scores in predicting psychiatric disorders has been documented with data showing that young children with lower competency were at risk for later social-emotional problems [29].

Longitudinal follow up of our cohort demonstrated continued moderate-high correlation of ASQ-SE at 2 years with CBCL1.5–5 at 48 months, thus reinforcing previously reported developmental continuity of psychopathology.

4.2. Maternal mental health and socio-emotional development in children

The importance of early childhood adversity and toxic stress and its role in socio-emotional development is being increasingly recognized. The central role of ecobiodevelopmental framework involving family, social relationships, environmental influences in the child's future socio-emotional and developmental trajectory is now accepted [3]. The AAP-Task Force on Mental Health recommends early identification approaches related to family factors and screening for maternal depression during well child visits [9,30].

Our study found a significant association of maternal mental health concerns with the child's socio-emotional development. Significantly higher maternal STAI and BDI-II scores were seen in children with "at risk" ASQ-SE scores and highlight the reported impact of maternal anxiety and depression in the first 2.5 years of a child's life on the mother-child dyad relationship, behavioral regulation and socio-emotional development [6,7,31].

A family-centered approach with timely identification of risk factors and intervention for maternal mental health concerns has been shown to mediate improvements in externalizing and internalizing problem behavior in children [32]. Our findings provide further support for serving the child-parent dyad as one entity for early identification and intervention [33].

5. Strengths and limitations

5.1. Strength

This prospective study was done in a low-risk, healthy population and thus findings can be generalized to match a primary care setting with potential for widespread population implementation. Our cohort size was large enough to generate robust cut-off scores. Furthermore, availability of data on maternal mental health status, from birth to 24 months of age facilitated identification of families at-risk rather than just children at-risk and can enable targeted dyadic mother-child interventions, and support for maternal mental health. With no current local data on socio-emotional screening of young children, this study fills an important gap in screening and identification of young toddlers at risk.

6. Limitations

Our study has limitations similar to other studies using the CBCL as a criterion measure in that both the focal and criterion measures viz. the ASQ-SE, and CBCL, are parent-completed measures. While ASQ-SE was shown to have acceptable psychometric properties at 24 months in our low-risk Singapore cohort its lower sensitivity may indicate that the use of additional screening instruments, DSM-keyed structured interviews or direct observational methods as criterion measures could also be needed to improve detection of children and families at-risk. Additionally, our study used the ASQ-SE which was the available tool at the time of the study. Further research with the current tool ASQ-SE 2 would be useful to enhance its validity.

We also acknowledge that the study sample may not be representative of the Singapore population, but we sampled from the two largest maternity hospitals in Singapore and recruited both private and subsidized patients.

7. Conclusion

The ASQ-SE with a local cut-off threshold had adequate psychometric properties at 24 months of age in our low-risk Singapore cohort, with good internal consistency and acceptable sensitivity and specificity. Our study also found significant relationships between maternal mental health concerns such as depression, anxiety, and the child's socioemotional competence.

Thus the ASQ-SE can be a useful, convenient screening tool for primary health care use in Singapore for socio-emotional competence with identification of relevant local cutoffs and risk factors of maternal mental health. This would be helpful in resource planning for national healthcare spending.

CRediT authorship contribution statement

Pratibha Keshav Agarwal: Writing – original draft, Writing – review & editing, Formal analysis. **Huichao Xie:** Formal analysis, Writing – review & editing. **Anu Sathyan Sathyapalan Rema:** Data curation, Formal analysis. **Ellen Ghim Hoon Tay:** Writing – review & editing. **Michael J. Meaney:** Writing – review & editing. **Keith M. Godfrey:** Writing – review & editing. **Shirong Cai:** Writing – review & editing. **Helen Yu Chen:** Writing – review & editing. **Yap Seng Chong:** Writing – review & editing. **Victor Samuel Rajadurai:** Writing – review & editing. **Lourdes Mary Daniel:** Conceptualization, Writing – review & editing.

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References

- [1] T.G. Halle, K.E. Darling-Churchill, Review of measures of social and emotional development, *J. Appl. Dev. Psychol.* 45 (2016) 8–18.
- [2] B. Daelmans, G.L. Darmstadt, J. Lombardi, M.M. Black, P.R. Britto, S. Lye, et al., Early childhood development: the foundation of sustainable development, *Lancet* 389 (10064) (2017) 9–11.
- [3] J.P. Shonkoff, A.S. Garner, The lifelong effects of early childhood adversity and toxic stress, *Pediatrics* 129 (1) (2012) e232–e246, <https://doi.org/10.1542/peds.2011-2663> (PubMed PMID: 22201156, Epub 20111226).
- [4] C. Weitzman, L. Wegner, Promoting optimal development: screening for behavioral and emotional problems, *Pediatric* 135 (2) (2015) 384–395, <https://doi.org/10.1542/peds.2014-3716> (PubMed PMID: 25624375).
- [5] M.M. Gleason, E. Goldson, M.W. Yogman, Addressing early childhood emotional and behavioral problems, *Pediatrics* 138 (6) (2016), <https://doi.org/10.1542/peds.2016-3025> (PubMed PMID: 27940734).
- [6] A. Rogers, S. Obst, S.J. Teague, L. Rossen, E.A. Spry, J.A. Macdonald, et al., Association between maternal perinatal depression and anxiety and child and adolescent development: a Meta-analysis, *JAMA Pediatr.* 174 (11) (2020) 1082–1092, <https://doi.org/10.1001/jamapediatrics.2020.2910> (PubMed PMID: 32926075; PubMed Central PMCID: PMC7490743).
- [7] B.A. Dachew, J.E. Heron, R. Alati, Parental depressive symptoms across the first three years of a child's life and emotional and behavioural problem trajectories in children and adolescents, *J Psychiatr Res.* 159 (2023) 135–144, <https://doi.org/10.1016/j.jpsychires.2023.01.019> (PubMed PMID: 36716565, Epub 20230113).
- [8] C.H. Zeanah, *Handbook of Infant Mental Health*, Guilford Publications, 2018.
- [9] J.M. Foy, Enhancing pediatric mental health care: report from the American Academy of Pediatrics task force on mental health, Introduction. *Pediatrics*. 125 (Suppl. 3) (2010) S69–S74, <https://doi.org/10.1542/peds.2010-0788C> (PubMed PMID: 20519564).
- [10] W. Gardner, K.J. Kelleher, K.A. Pajer, J.V. Campo, Primary care clinicians' use of standardized tools to assess child psychosocial problems, *Ambul. Pediatr.* 3 (4) (2003) 191–195, [https://doi.org/10.1367/1539-4409\(2003\)003<0191:pcuos>2.0.co;2](https://doi.org/10.1367/1539-4409(2003)003<0191:pcuos>2.0.co;2) (PubMed PMID: 12882596).
- [11] P.H. Lipkin, M.M. Macias, K.W. Norwood, T.J. Brei, L.F. Davidson, B.E. Davis, et al., Promoting optimal development: identifying infants and young children with developmental disorders through developmental surveillance and screening, *Pediatrics* 145 (1) (2020).
- [12] M. Pontoppidan, N.K. Niss, J.H. Pejtersen, M.M. Julian, M.S. Væver, Parent report measures of infant and toddler social-emotional development: a systematic review, *Fam. Pract.* 34 (2) (2017) 127–137, <https://doi.org/10.1093/fampra/cmx003> (PubMed PMID: 28158522).
- [13] T.M. Achenbach, L.A. Rescorla, *Manual for the ASEBA Preschool Forms and Profiles*, University of Vermont, Research center for children, youth ..., Burlington, VT, 2000.
- [14] J. Squires, D. Bricker, E. Twombly, S. Yockelson, M.S. Davis, Y. Kim, *Ages & Stages Questionnaires: Social-Emotional (ASQ: SE): A Parent-Completed, Child-Monitoring System for Social-Emotional Behaviors*, Paul H. Brookes Publishing Company Baltimore, MD, 2002.
- [15] J. Squires, D. Bricker, K. Heo, E. Twombly, Identification of social-emotional problems in young children using a parent-completed screening measure, *Early Child. Res. Q.* 16 (4) (2001) 405–419.
- [16] C.-Y. Chieh, J. Squires, C.-I. Chen, R. Wu, H. Xie, The adaptation and psychometric examination of a social-emotional developmental screening tool in Taiwan, *Early Educ. Dev.* 31 (1) (2020) 27–46.
- [17] H. Xie, N. Waschl, X. Bian, R. Wang, C.-Y. Chen, L. Anunciação, Z. Chai, W. Song, Y. Li, Validity studies of a parent completed social-emotional measure in a representative sample of young children in China, *Appl. Dev. Sci.* 26 (4) (2021) 689–703, <https://doi.org/10.1080/10888691.2021.1977642>.
- [18] H. Wang, J. Zhang, X. Huang, G. Liu, G. Lian, S. Shi, 中国城市幼儿情绪及社会性发展量表标准化的信度和效度分析 [Reliability and validity of the standardized Chinese version of the urban infant-toddler social and emotional assessment (CITSEA)], *Chinese Journal of Child Health Care* 17 (3) (2009) 271–274.
- [19] L. Liu, L. Wu, K. Yao, 2 ~ 3 岁幼儿行为量表全国城市常模的制定 [Institution of child behavior checklist (CBCL) norm for 2 to 3 years children in national cities], *Chinese Journal of Child Health Care* 11 (6) (2003) 377–379.
- [20] S.-E. Soh, M.T. Tint, P.D. Gluckman, K.M. Godfrey, A. Rifkin-Graboy, Y.H. Chan, et al., Cohort profile: growing up in Singapore towards healthy outcomes (GUSTO) birth cohort study, *Int. J. Epidemiol.* 43 (5) (2014) 1401–1409.
- [21] H. Xie, J. Clifford, J. Squires, C.-Y. Chen, X. Bian, Q. Yu, Adapting and validating a developmental assessment for chinese infants and toddlers: the ages & stages questionnaires: inventory, *Infant Behav. Dev.* 49 (2017) 281–295.
- [22] C.D. Spielberger, *State-Trait Anxiety Inventory for Adults*, 1983.
- [23] A.T. Beck, R.A. Steer, G. Brown, *Beck depression inventory-II (BDI-II)*, APA Psych Tests, 1996, <https://doi.org/10.1037/t00742-000>.
- [24] L. Alvarez-Núñez, M. González, F. Rudnitsky, A. Vázquez-Echeverría, Psychometric properties of the Spanish version of the Ages & Stages Questionnaires: Social-Emotional in a nationally representative sample, *Early Hum. Dev.* 149 (2020) 105157.
- [25] K. Stensen, F. Stenseng, S. Lydersen, M.S. de Wolff, J. Wallander, M.B. Drugli, Screening for mental health problems in a Norwegian preschool population. A validation of the ages and stages questionnaire: social-emotional (ASQ: SE), *Child Adolesc. Mental Health* 23 (4) (2018) 368–375.
- [26] M.S. de Wolff, M.H. Theunissen, A.G. Vogels, S.A. Reijneveld, Three questionnaires to detect psychosocial problems in toddlers: a comparison of the BITSEA, ASQ: SE, and KIPPP, *Acad. Pediatr.* 13 (6) (2013) 587–592.
- [27] T. Velikonja, J. Edbrooke-Childs, A. Calderon, M. Slead, A. Brown, J. Deighton, The psychometric properties of the ages & stages questionnaires for ages 2-2.5: a systematic review, *Child Care Health Dev.* 43 (1) (2017) 1–17.
- [28] J.V. Lavigne, K.M. Meyers, M. Feldman, Systematic review: classification accuracy of behavioral screening measures for use in integrated primary care settings, *J. Pediatr. Psychol.* 41 (10) (2016) 1091–1109.
- [29] M.J. Briggs-Gowan, A.S. Carter, Social-emotional screening status in early childhood predicts elementary school outcomes, *Pediatrics* 121 (5) (2008) 957–962, <https://doi.org/10.1542/peds.2007-1948> (PubMed PMID: 18450899; PubMed Central PMCID: PMC3088107).
- [30] *Pediatrics AAO, Addressing Mental Health Concerns in Primary Care: A clinician's Toolkit*, Elk Grove Village, IL, American Academy of Pediatrics, 2010.
- [31] T.N. Chorbadian, A. Deavenport-Saman, C. Higgins, S.M. Chao, J.H. Yang, I. Koolwijk, et al., Maternal depressive symptoms and developmental delay at age 2: a diverse population-based longitudinal study, *Matern. Child Health J.* 24 (2020) 1267–1277.
- [32] D.S. Shaw, A. Connell, T.J. Dishion, M.N. Wilson, F. Gardner, Improvements in maternal depression as a mediator of intervention effects on early childhood problem behavior, *Dev. Psychopathol.* 21 (2) (2009) 417–439.
- [33] C.J. Dunst, C.M. Trivette, D.W. Hamby, Meta-analysis of family-centered helping practices research, *Ment. Retard. Dev. Disabil. Res. Rev.* 13 (4) (2007) 370–378.